A METHOD FOR ASYNCHRONOUS SECURE COMMUNICATION WITH POST-HOC KEY DERIVATION

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Abstract

A method for secure communication that inverts the traditional cryptographic paradigm to significantly increase resilience against man-in-the-middle attacks. The method comprises transmitting an encrypted message, M, encrypted with a first key, K1, that is never transmitted. Subsequently, a prefix, P, and a suffix, S, are transmitted independently. A second key, K2, is then derived from K1 and the full concatenated data block (P+M+S). Only after these transmissions are complete is K2 transmitted to the recipient. An attacker must successfully intercept four separate and asynchronous transmissions, potentially across different channels, and correctly associate them to compromise the communication.

Background of the Invention

Conventional secure communication protocols, from Transport Layer Security (TLS) to public-key cryptography, operate on a sequential, "key-first" paradigm. A secure chan-

nel is established, cryptographic keys are exchanged (e.g., via Diffie-Hellman or RSA), and only then is the encrypted data transmitted. This model's primary vulnerability is the key exchange itself. An attacker who can intercept the key exchange in real-time (a "Man-in-the-Middle" or MITM attack) can often compromise the entire session that follows. The success of such an attack is predicated on the temporal and spatial collision of the attacker with a single, critical event: the key exchange. There is a need for a communication method that is not vulnerable to a single-point-of-failure interception.

Summary of the Invention

The present invention provides a method for secure communication that decouples the transmission of the data from the transmission of the final cryptographic key, providing "backwards secrecy." Instead of sending the key first, the data is sent in multiple parts, and the key required for decryption is the last element to be transmitted.

The process begins with a sender generating a first, secret key (K1) which is never transmitted. The core message (M) is encrypted with K1 and transmitted to a receiver. At a subsequent time, and optionally over a different communication channel, the sender transmits a prefix (P). At another subsequent time, the sender transmits a suffix (S). With the three data components now transmitted, the sender computes a second, final key (K2) based on K1 and the fully assembled data block (P+M+S). Finally, at another subsequent time, the sender transmits K2 to the receiver. The receiver assembles the data block and uses the received K2 to decrypt it. This method drastically increases the difficulty for an attacker, who must now intercept four distinct transmissions and correctly identify their relationship without any contextual clues, and crucially, the transmitted key (K2) is only viable to decrypt the fully assembled data block (P+M+S) but not P, M, or S individually, or in other words, the message transmitted (M encrypted using key K1) cannot be decrypted with the key transmitted (K2).

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the sequence of transmissions and key derivation steps of the secure communication method, showing the asynchronous interactions between the sender and receiver.

Detailed Description of the Invention

Referring to **FIG. 1**, the method for asynchronous secure communication is illustrated. The process is divided between a Sender (101) and a Receiver (102).

The process begins at the Sender (101). In step (110), an initial, internal-only secret key, K1, is generated. This key is never transmitted in any form. In step (112), the core message, M, is encrypted using a symmetric encryption algorithm with K1 as the key.

In step (114), the encrypted message M is transmitted to the Receiver (102), who receives and stores it (150). Subsequently, in steps (116) and (118), the Sender transmits a prefix P and a suffix S, respectively. These transmissions are asynchronous, meaning they can occur at different times and optionally over different communication channels (e.g., M over satellite, P over fiber, S via a dead drop). The Receiver (102) collects and stores P and S as they arrive (152, 154).

After the three data components have been sent, the Sender (101) proceeds to step (120). The Sender concatenates the transmitted components in the correct order to form a final message block (P+M+S). A second key, K2, is then computed using a deterministic process, for example, by using a Key Derivation Function (KDF) that takes K1 and a cryptographic hash of the final message block as input (e.g., K2 = KDF(K1, SHA-256(P+M+S))).

Finally, in step (122), the Sender transmits the final key, K2. The Receiver (102) receives K2 (156). With all four components now received, the Receiver proceeds to the final step (158). The Receiver assembles the message block (P+M+S) in the correct order and uses the received key K2 to decrypt it, recovering the original plaintext message.

Claims

What is claimed is:

1. A method for secure communication between a sender and a receiver, comprising the steps of:
    - at the sender, generating a first secret key, K1, which is not transmitted to the receiver;
    - at the sender, encrypting a message M using K1 to produce an encrypted message;

- transmitting the encrypted message from the sender to the receiver;
- transmitting a prefix data block P from the sender to the receiver;
- transmitting a suffix data block S from the sender to the receiver;
- at the sender, deriving a second key, K2, from K1 and the combination of P, the encrypted message, and S;
- transmitting K2 from the sender to the receiver; and
- at the receiver, decrypting the combination of P, the encrypted message, and S using K2.

2. The method of claim 1, wherein the encrypted message, the prefix P, the suffix S, and the second key K2 are transmitted at different points in time.

3. The method of claim 1, wherein at least two of the transmissions from the group consisting of the encrypted message, the prefix P, the suffix S, and the second key K2, are performed over different communication channels.

4. The method of claim 1, wherein the step of deriving K2 comprises computing a cryptographic hash of the concatenation of P, the encrypted message, and S, and using the hash and K1 as inputs to a key derivation function.

5. A system for secure communication, comprising:

   - a sender device configured to:
     - generate a non-transmitted first key K1;
     - encrypt a message M with K1;
     - transmit the encrypted M, a prefix P, and a suffix S in separate transmissions;
     - derive a second key K2 from K1 and the combination of P, M, and S; and
     - transmit K2;
   - a receiver device configured to:
     - receive the encrypted M, P, S, and K2;
     - assemble the received P, M, and S; and
     - decrypt the assembled data using K2.

6. The system of claim 5, wherein the sender device is configured to perform the transmissions asynchronously.

Drawings

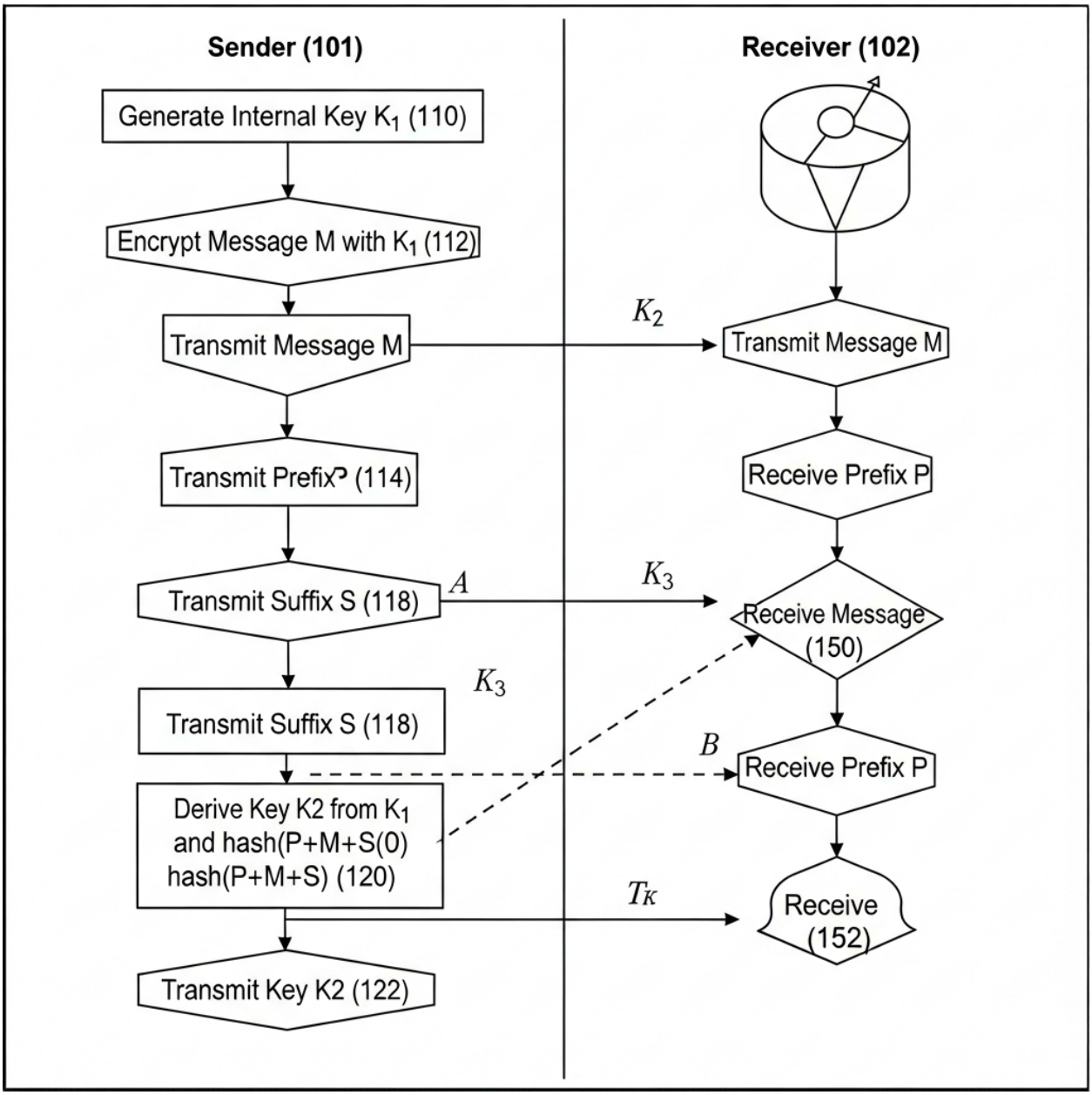


Figure 1: Flowchart illustrating the sequence of transmissions and key derivation steps of the secure communication method.